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## ARRAYS

**Aim:**

To understand and implement array operations in Java.

**PRE LAB EXERCISE****QUESTIONS**

- ✓ What is an array?

An array is a collection of elements of the same data type stored in continuous memory locations.

Each element is accessed using an index number.

Example:

```
int[] a = {10, 20, 30, 40};
```

- ✓ Why are arrays used?

Arrays are used because:

- They store multiple values using a single variable name
- They provide easy access to elements using index
- They make data processing faster
- They help in implementing sorting, searching, matrices, etc.

- ✓ What is the difference between array and variable?

| Variable                          | Array                                       |
|-----------------------------------|---------------------------------------------|
| Stores only one value             | Stores multiple values                      |
| Holds a single data item          | Holds a collection of items                 |
| Example: <code>int x = 10;</code> | Example: <code>int[] x = new int[5];</code> |
| Simple memory usage               | Uses continuous memory                      |

**IN LAB EXERCISE****Objective:**

To perform array operations using simple programs.

**PROGRAMS:**

## 1. Program to Read and Print Array Elements

### Code:

```
import java.util.Scanner;

public class ReadPrintArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        System.out.println("Array elements are:");

        for(int i = 0; i < 5; i++)

            System.out.print(arr[i] + " ");

    }

}
```

### OUTPUT:

#### Input:

10 20 30 40 50

#### Output:

Array elements are:

10 20 30 40 50

```
2 import java.util.Scanner;
3 public class ReadPrintArray {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         System.out.println("Enter 5 elements:");
8         for(int i = 0; i < 5; i++)
9             arr[i] = sc.nextInt();
10        System.out.println("Array elements are:");
11        for(int i = 0; i < 5; i++)
12            System.out.print(arr[i] + " ");
13    }
14 }
15
```

Problems @ Javadoc Declaration Console X Eclipse IDE for Java  
<terminated> ReadPrintArray [Java Application] C:\Users\sanji\Downloads\eclipse-jav  
Enter 5 elements:  
10 20 30 40 50  
Array elements are:  
10 20 30 40 50

## 2. Program to Find Sum of Array Elements

### Code:

```
import java.util.Scanner;

public class SumArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int sum = 0;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++)

            sum += arr[i];

        System.out.println("Sum = " + sum);

    }

}
```

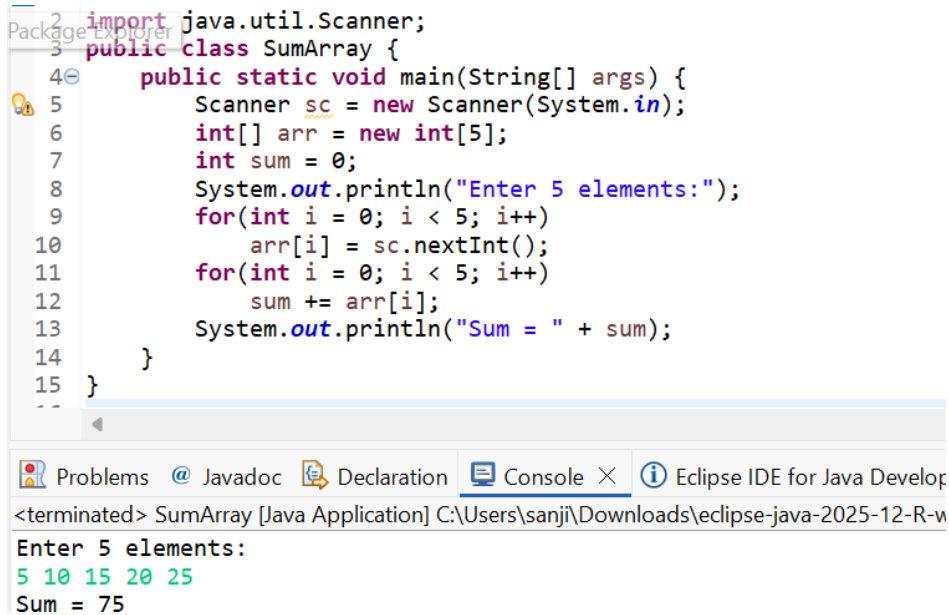
### OUTPUT:

**Input:**

5 10 15 20 25

**Output:**

Sum = 75



```
1 import java.util.Scanner;
2
3 public class SumArray {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         int sum = 0;
8         System.out.println("Enter 5 elements:");
9         for(int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11         for(int i = 0; i < 5; i++)
12             sum += arr[i];
13         System.out.println("Sum = " + sum);
14     }
15 }
```

<terminated> SumArray [Java Application] C:\Users\sanji\Downloads\eclipse-java-2025-12-R-v

Enter 5 elements:  
5 10 15 20 25  
Sum = 75

**3. Program to Find Largest Element in an Array****Code:**

```
import java.util.Scanner;

public class LargestElement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        int max = arr[0];

        for(int i = 1; i < 5; i++)

            if(arr[i] > max)

                max = arr[i];

    }

}
```

```
        System.out.println("Largest element = " + max);
    }
}
```

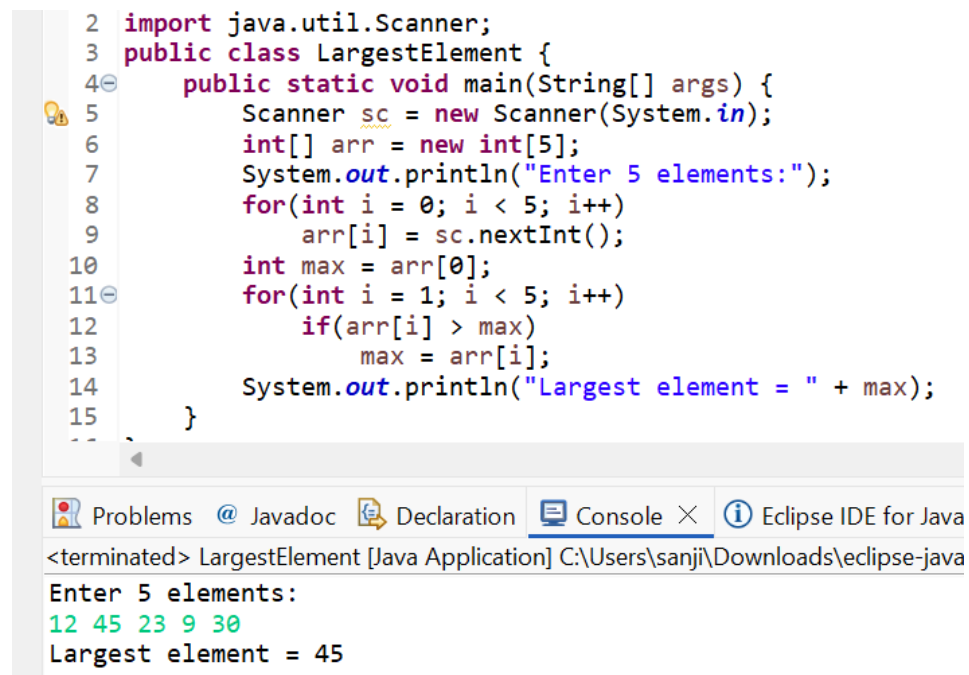
## OUTPUT:

### Input:

12 45 23 9 30

### Output:

Largest element = 45



```
2 import java.util.Scanner;
3 public class LargestElement {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         System.out.println("Enter 5 elements:");
8         for(int i = 0; i < 5; i++)
9             arr[i] = sc.nextInt();
10        int max = arr[0];
11        for(int i = 1; i < 5; i++)
12            if(arr[i] > max)
13                max = arr[i];
14        System.out.println("Largest element = " + max);
15    }
16 }
```

Problems @ Javadoc Declaration Console × Eclipse IDE for Java

<terminated> LargestElement [Java Application] C:\Users\sanji\Downloads\eclipse-java

Enter 5 elements:  
12 45 23 9 30  
Largest element = 45

## 4. Program to Reverse an Array

### Code:

```
import java.util.Scanner;
```

```
public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
```

```

        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();

        System.out.println("Reversed array:");

        for(int i = 4; i >= 0; i--)
            System.out.print(arr[i] + " ");

    }
}

```

## OUTPUT:

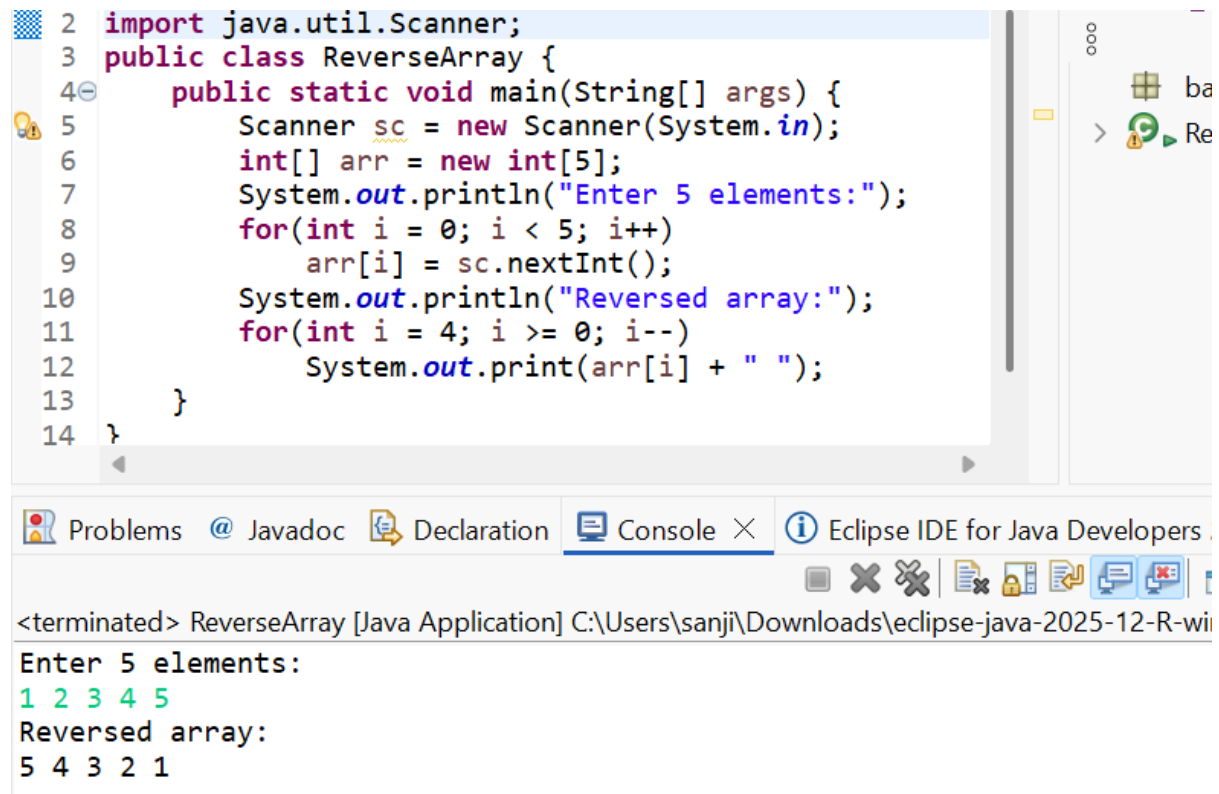
### Input:

1 2 3 4 5

### Output:

Reversed array:

5 4 3 2 1



```

2 import java.util.Scanner;
3 public class ReverseArray {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         System.out.println("Enter 5 elements:");
8         for(int i = 0; i < 5; i++)
9             arr[i] = sc.nextInt();
10        System.out.println("Reversed array:");
11        for(int i = 4; i >= 0; i--)
12            System.out.print(arr[i] + " ");
13    }
14 }

```

Problems Javadoc Declaration Console Eclipse IDE for Java Developers

<terminated> ReverseArray [Java Application] C:\Users\sanji\Downloads\eclipse-java-2025-12-R-wi

Enter 5 elements:  
1 2 3 4 5  
Reversed array:  
5 4 3 2 1

## 5. Program to Count Even and Odd Numbers

### Code:

```
import java.util.Scanner;

public class EvenOddCount {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int even = 0, odd = 0;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++) {

            if(arr[i] % 2 == 0)

                even++;

            else

                odd++;

        }

        System.out.println("Even = " + even);

        System.out.println("Odd = " + odd);

    }

}
```

### **OUTPUT:**

#### **Input:**

2 7 4 9 10

#### **Output:**

Even = 3

Odd = 2

```
2 import java.util.Scanner;
3 public class EvenOddCount {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         int even = 0, odd = 0;
8         System.out.println("Enter 5 elements:");
9         for(int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11         for(int i = 0; i < 5; i++) {
12             if(arr[i] % 2 == 0)
13                 even++;
14             else
15                 odd++;
16         }
17
18         System.out.println("Even = " + even);
19         System.out.println("Odd = " + odd);
20     }
}
```

Problems @ Javadoc Declaration Console × Eclipse IDE for Java [

<terminated> EvenOddCount [Java Application] C:\Users\sanji\Downloads\eclipse-java-

Enter 5 elements:

2 7 4 9 10

Even = 3

Odd = 2

## 6. Program to Sort Array in Ascending Order

### Code:

```
import java.util.Scanner;

public class SortArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int temp;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++) {
```



```
        for(int j = i + 1; j < 5; j++) {  
            if(arr[i] > arr[j]) {  
                temp = arr[i];  
                arr[i] = arr[j];  
                arr[j] = temp;  
            }  
        }  
    }  
    System.out.println("Sorted array:");  
    for(int i = 0; i < 5; i++)  
        System.out.print(arr[i] + " ");  
    }  
}
```

### **OUTPUT:**

#### **Input:**

45 12 78 23 9

#### **Output:**

Sorted array:

9 12 23 45 78

```
2 import java.util.Scanner;
3 public class SortArray {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         int temp;
8         System.out.println("Enter 5 elements:");
9         for(int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11         for(int i = 0; i < 5; i++) {
12             for(int j = i + 1; j < 5; j++) {
13                 if(arr[i] > arr[j]) {
14                     temp = arr[i];
15                     arr[i] = arr[j];
16                     arr[j] = temp;
17                 }
18             }
19         }
20         System.out.println("Sorted array:");
21         for(int i = 0; i < 5; i++)
22             System.out.print(arr[i] + " ");
23     }
24 }
25
```

Problems @ Javadoc Declaration Console X Eclipse IDE for Java Developers 2026-03 M

<terminated> SortArray [Java Application] C:\Users\sanji\Downloads\eclipse-java-2025-12-R-win32-x86\_64\

Enter 5 elements:

45 12 78 23 9

Sorted array:

9 12 23 45 78

## 7. Program to Find Second Largest Element

### Code:

```
import java.util.Scanner;
```

```
public class SecondLargest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");

        for (int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();
```

```
int largest = arr[0];
int second = Integer.MIN_VALUE;

for (int i = 1; i < 5; i++) {
    if (arr[i] > largest) {
        second = largest;
        largest = arr[i];
    }
    else if (arr[i] > second && arr[i] != largest) {
        second = arr[i];
    }
}

System.out.println("Second largest = " + second);
}
```

### **OUTPUT:**

#### **Input:**

10 45 23 89 67

#### **Output:**

Second largest = 67

```
2 import java.util.Scanner;
3 public class SecondLargest {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7
8         System.out.println("Enter 5 elements:");
9         for (int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11
12         int largest = arr[0];
13         int second = Integer.MIN_VALUE;
14
15         for (int i = 1; i < 5; i++) {
16             if (arr[i] > largest) {
17                 second = largest;
18                 largest = arr[i];
19             }
20             else if (arr[i] > second && arr[i] != largest) {
21                 second = arr[i];
22             }
23         }
24
25         System.out.println("Second largest = " + second);
26     }
}
```

Problems Javadoc Declaration Console Eclipse IDE for Java Developers 20

<terminated> SecondLargest [Java Application] C:\Users\sanji\Downloads\eclipse-java-2025-12-R-wir

Enter 5 elements:  
10 45 23 89 67  
Second largest = 67

## 8. Program for Matrix Addition (2D Array)

### Code:

```
import java.util.Scanner;

public class MatrixAddition {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[][] a = new int[2][2];
        int[][] b = new int[2][2];
        int[][] sum = new int[2][2];

        System.out.println("Enter elements of matrix A:");

        for(int i = 0; i < 2; i++)

            for(int j = 0; j < 2; j++)

                a[i][j] = sc.nextInt();
```

```

System.out.println("Enter elements of matrix B:");
for(int i = 0; i < 2; i++)
    for(int j = 0; j < 2; j++)
        b[i][j] = sc.nextInt();
for(int i = 0; i < 2; i++)
    for(int j = 0; j < 2; j++)
        sum[i][j] = a[i][j] + b[i][j];
System.out.println("Sum matrix:");
for(int i = 0; i < 2; i++) {
    for(int j = 0; j < 2; j++)
        System.out.print(sum[i][j] + " ");
    System.out.println();
}
}
}

```

### **OUTPUT:**

Matrix A:

1 2

3 4

Matrix B:

5 6

7 8

### **Sum matrix:**

6 8

10 12

```
2 import java.util.Scanner;
3 public class MatrixAddition {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[][] a = new int[2][2];
7         int[][] b = new int[2][2];
8         int[][] sum = new int[2][2];
9         System.out.println("Enter elements of matrix A:");
10        for(int i = 0; i < 2; i++)
11            for(int j = 0; j < 2; j++)
12                a[i][j] = sc.nextInt();
13        System.out.println("Enter elements of matrix B:");
14        for(int i = 0; i < 2; i++)
15            for(int j = 0; j < 2; j++)
16                b[i][j] = sc.nextInt();
17        for(int i = 0; i < 2; i++)
18            for(int j = 0; j < 2; j++)
19                sum[i][j] = a[i][j] + b[i][j];
20        System.out.println("Sum matrix:");
21        for(int i = 0; i < 2; i++) {
22            for(int j = 0; j < 2; j++)
23                System.out.print(sum[i][j] + " ");
24            System.out.println();
25        }
26    }
}
```

Problems @ Javadoc Declaration Console Eclipse IDE for Java Developers 2026-03 M2

<terminated> MatrixAddition [Java Application] C:\Users\sanji\Downloads\eclipse-java-2025-12-R-win32-x86\_64\eclipse\pl

Enter elements of matrix A:

1 2  
3 4

Enter elements of matrix B:

<terminated> MatrixAddition [Java Application] C:\Users\sanji\Downloads\eclipse-java-2025-

Enter elements of matrix A:

1 2  
3 4

Enter elements of matrix B:

5 6  
7 8

Sum matrix:

6 8  
10 12

## POST LAB EXERCISE

✓ Why is array indexing usually started from zero instead of one?

Array indexing usually started from zero because:

- Arrays are stored in memory blocks
- The index represents the offset from the base address
- First element is at base address + 0
- This improves performance and memory calculation

Example:

`arr[0]` refers to the first element

- ✓ What happens if we try to access an array element outside its declared size?

It causes a runtime error.

In Java, it throws:

`ArrayIndexOutOfBoundsException`

Example:

```
int[] a = new int[5];
```

```
System.out.println(a[6]); // Error
```

- ✓ How does memory allocation differ for static arrays and dynamic arrays?

| Static Array                    | Dynamic Array                                                    |
|---------------------------------|------------------------------------------------------------------|
| Size is fixed at compile time   | Size can be decided at runtime                                   |
| Cannot change size              | Can change size (using collections like <code>ArrayList</code> ) |
| Faster access                   | Slightly slower                                                  |
| Example: <code>int a[5];</code> | Example: <code>new int[n];</code>                                |

- ✓ Why is searching faster in arrays compared to linked lists?

Searching faster in arrays than linked lists because:

- Arrays support direct access using index
- Time complexity is  $O(1)$  for accessing elements
- Linked lists require traversing node by node
- Hence, searching in arrays is faster

- ✓ What is the difference between contiguous and non-contiguous memory allocation?

| Contiguous Memory            | Non-Contiguous Memory       |
|------------------------------|-----------------------------|
| Memory blocks are continuous | Memory blocks are scattered |
| Used in arrays               | Used in linked lists        |
| Faster access                | Slower access               |
| Easy indexing                | No direct indexing          |

**Result:**

Thus the array operations were executed successfully.

**ASSESSMENT**

| Description       | Max Marks | Marks Awarded |
|-------------------|-----------|---------------|
| Pre Lab Exercise  | 5         |               |
| In Lab Exercise   | 10        |               |
| Post Lab Exercise | 5         |               |
| Viva              | 10        |               |
| Total             | 30        |               |
| Faculty Signature |           |               |